

Git & GitHub Cheat Sheet

1. Git vs GitHub

- Git: Distributed version control system; tracks code changes locally.
- GitHub: Cloud platform to host Git repositories, enabling collaboration.
- Memory Tip: Git = notebook, GitHub = Google Drive for your notebook.

2. Key Concepts

- Commit: Snapshot of staged changes; tracks history ("Save point")
- Branch: Separate line of development ("Independent workspace")
- Staged Changes: Changes ready to commit ("Included in photo")
- Unstaged Changes: Changes not yet staged ("Things in the room")
- Clone: Copy of a repo locally ("Download workspace")
- Fork: Online copy on GitHub ("Personal copy of repo")
- Pull Request: Request to merge your branch for review ("Please review and merge my changes")

3. Basic Git Commands

- Initialize repo: `git init`
- Check branch: `git branch`
- Stage changes: `git add <file>`
- Commit changes: `git commit -m "msg"`
- Push to remote: `git push origin <branch>`
- Pull updates: `git pull origin <branch>`
- Undo last commit (keep changes): `git reset --soft HEAD~1`
- Merge branch: `git checkout main + git merge feature-branch`

4. GitHub Workflow

1. Fork repo -> create your copy on GitHub
2. Clone fork -> copy to local machine
3. Create branch -> `git checkout -b feature-x`
4. Make changes, stage, commit -> `git add . + git commit -m "msg"`

5. Push branch -> git push origin feature-x

6. Open Pull Request -> merge changes to main after review

5. Linking Local Repo to GitHub

- HTTPS: git remote add origin https://github.com/username/repo.git
- SSH: git remote add origin git@github.com:username/repo.git
- Verify: git remote -v

6. Why Protect Main Branch

- Ensures stable, deployable code
- Developers work in separate branches and merge only after review/tests
- Prevents accidental breaks in production

Memory Tip Summary

- Commit = save point / snapshot
- Branch = workspace / feature line
- Staged = ready for commit
- Unstaged = changes not yet staged
- Clone = local copy
- Fork = personal online copy
- Pull request = request review & merge